

PROTOCOL

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Impact of cycling exercise in patients with chronic kidney disease: protocol for a systematic review and trial sequential meta-analysis

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Abstract

Background Chronic kidney disease is a global public health problem affecting approximately 10% of the adult population. Conventional management combines pharmacotherapy and dialysis, yet long-term complications persist. Cycling, a low joint load exercise, may improve cardiopulmonary function and renal outcomes, but conflicting evidence exists regarding its efficacy in advanced CKD.

Methods and analysis Randomized controlled trials (RCTs) that compare cycling exercise and usual care in patients with chronic kidney disease will be included. Literature searches will be conducted in PubMed, Web of Science, Embase, and Cochrane Library. Two reviewers will independently perform the processes of literature retrieval, screening, data extraction, and assessment of risk of bias. Risk of bias in included studies will be evaluated using Revised Cochrane risk-of-bias tool (ROB 2) for RCTs. Review Manager (RevMan) will be used for data pooling. Subgroup analysis, trial sequential analysis (TSA), and sensitivity analysis will be conducted.

Ethics and dissemination Ethical approval is not required because this study is a secondary analysis of existing data. We will disseminate the findings through peer-reviewed publications.

Systematic review registration PROSPERO CRD420251048364.

Keywords Cycling, Chronic kidney disease, CKD, Exercise

Introduction

Chronic kidney disease (CKD) is a global public health problem affecting approximately 10% of the adult population [1]. CKD is characterized by a progressive decline in renal function, which may ultimately lead to end-stage

renal disease, with dependence on dialysis or renal transplantation [2, 3]. This burden of CKD has escalated alongside global aging populations and the rising prevalence of diabetes and hypertension, which may impose substantial strain on the healthcare system and the socio-economy [4, 5]. In addition to the deterioration of renal function, patients with CKD are often accompanied by complications such as cardiovascular disease, muscle atrophy, metabolic disorders, and immune deficiencies, leading to a significant reduction in quality of life and an increase in mortality [6–9].

Traditional CKD management primarily relies on pharmacological control and renal replacement therapy

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(dialysis and renal transplantation). For patients who have progressed to end-stage renal diseases, dialysis treatment is essential to sustain life, effectively replacing some renal functions and alleviating uremic symptoms [10–12]. However, long-term dialysis patients are often at increased risk of cardiovascular complications, accelerated muscle wasting, and reduced physical activity capacity. In addition, dialysis treatment requires frequent visits to medical facilities, potentially restricting engagement in daily activities [13, 14]. Emerging evidence underscores non-pharmacological interventions, particularly exercise therapy, as a promising therapeutic strategy for its multi-dimensional benefits. Studies suggest that regular aerobic exercise can improve cardiopulmonary function, reduce chronic inflammation, and improve insulin sensitivity in patients with CKD, including those on dialysis [15, 16]. However, dialysis-related factors such as fatigue, anemia, and fluid overload frequently compromise exercise tolerance, necessitating tailored modalities that balance safety and efficacy. Cycling, as a low joint load exercise, can be performed in conjunction with dialysis treatment (e.g., cycling training during dialysis) or as a rehabilitation activity on non-dialysis days, and its clinical applicability is worthy of in-depth exploration [17–19].

Cycling has unique physiological and practical advantages. Cycling reduces joint loading and is particularly suitable for patients with obesity or osteoarthritis. Some clinical trials have shown that regular cycling can significantly improve muscle metabolic indexes in CKD patients and may even regulate oxidative stress and renal blood flow [20–22]. However, it has also been suggested that patients with CKD may not benefit from cycling interventions due to low fatigue tolerance and poor exercise compliance [23, 24]. Such contradictory results may be closely related to study design heterogeneity and population characteristics (e.g., CKD stage, comorbidities), and systematic evidence integration is urgently needed.

Based on the above background, this study intends to comprehensively assess the effects of cycling exercise on patients with CKD (including dialysis and non-dialysis populations) through systematic evaluation and trial sequential meta-analysis, and to clarify its clinical effectiveness and safety. The results of the review will provide high-quality evidence to support the development of CKD exercise rehabilitation guidelines.

Methods

Protocol and registration

In strict accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocols (PRISMA-P) guidelines, this systematic review and meta-analysis will be rigorously conducted. Prior to commencement, the study protocol was prospectively

registered with the PROSPERO international database (Registration ID: CRD420251048364), ensuring methodological transparency and scientific rigor [25].

Inclusion criteria

Participants

Adult patients (≥ 18 years) with chronic kidney disease.

Intervention

Supervised or unsupervised cycling exercise (e.g., stationary cycling during hemodialysis, home-based cycling), with sessions lasting ≥ 10 min per episode, at any frequency/intensity.

Comparison

Patients in the control group only received usual care.

Primary outcome

The primary outcome is 6-min walking distance (m).

Secondary outcomes

BMI; SF-36 Scale Scores.

Study design

We will include parallel-group randomized controlled trials. Crossover and cluster-RCT designs are excluded due to potential carryover effects and unit-of-analysis issues, respectively.

Exclusion criteria

The exclusion criteria are as follows: non-randomized studies (e.g., observational studies, case reports, meeting abstracts); pediatric or adolescent populations; participation in clinical studies during the 3 months prior; unwilling to participate in the trial; crossover studies will be excluded from this systematic review and meta-analysis.

Literature sources and retrieval strategy

To ensure methodological rigor and minimize bias, two reviewers will independently search four major electronic databases: PubMed, Web of Science, Embase, and the Cochrane Library. Databases will be searched from inception to July 31, 2025. Only English-language studies will be included. Discrepancies in study identification or selection will be resolved through discussion or adjudication by a third reviewer if consensus cannot be reached. The detailed retrieval strategy is shown in Table 1.

Literature screening and data extraction

To ensure methodological rigor and reproducibility, two reviewers will conduct literature screening and data extraction in strict accordance with PRISMA guidelines. Screening involves (1) title/abstract review and (2)

Table 1 The retrieval strategy

Search	Query
#1	((cycle[Title/Abstract]) OR (bicycle[Title/Abstract])) OR (bike[Title/Abstract])
#2	((chronic kidney disease[Title/Abstract]) OR (end-stage renal disease[Title/Abstract]) OR (dialysis[Title/Abstract])) OR (hemodialysis[Title/Abstract])
#3	((random[Title/Abstract]) OR (randomized controlled trial[Title/Abstract])) OR (RCT[Title/Abstract])
#4	((cycle[Title/Abstract]) OR (bicycle[Title/Abstract])) OR (bike[Title/Abstract]) AND (((chronic kidney disease[Title/Abstract]) OR (end-stage renal disease[Title/Abstract]) OR (dialysis[Title/Abstract])) OR (hemodialysis[Title/Abstract])) AND (((random[Title/Abstract]) OR (randomized controlled trial[Title/Abstract])) OR (RCT[Title/Abstract]))

full-text assessment, with discrepancies resolved through consensus or third-reviewer adjudication. The screening process will be reported using a PRISMA flow diagram. A standardized, piloted form will extract study design, participant characteristics, intervention details, and outcomes. Two reviewers will independently extract data, with disagreements resolved by a third reviewer.

Assessment of risk of bias

To ensure standardized and unbiased quality appraisal, two independent reviewers will assess the risk of bias of included randomized controlled trials using the Revised Cochrane risk-of-bias tool (ROB 2). This tool evaluates five critical domains: bias arising from the randomization process, deviations from intended interventions, missing outcome data, outcome measurements, and selection of the reported result. Controversies will be discussed and solved in the research team.

Statistical analysis

Data synthesis and statistical analyses will be performed using Review Manager software (version 5.4). For dichotomous variables, risk ratios (RR) with 95% confidence intervals (CI) will be derived using the Mantel–Haenszel method. Continuous variables will be pooled and analyzed as mean difference (MD) with 95% CIs, employing the inverse variance method. The statistical significance level (α) is set at 0.05. Statistical differences are deemed significant if the P value is less than 0.05. Random-effects models will be used for all meta-analyses due to anticipated clinical/methodological heterogeneity. Heterogeneity will be quantified using I^2 , with values of 0–30% considered low, 30–60% moderate, and > 60% substantial. Publication bias will be assessed using funnel plots and Egger’s regression test if ≥ 10 studies are included.

Subgroup analysis and sensitivity analysis

Subgroup analyses will be stratified by age (< 60 vs. ≥ 60 years) and dialysis status. Besides, sensitivity

analysis of the primary outcome will be conducted by excluding studies with high ROB or > 20% attrition.

Trial sequential analysis

Random errors can lead to spurious conclusions when a meta-analysis involves a limited number of trials and a small patient population. TSA will be employed to assess the risks associated with random errors due to insufficient sample size or repeated testing, and to estimate the required information size (RIS) for the meta-analysis. TSA will be conducted only if heterogeneity (I^2) is < 50% using the TSA version 0.9.5.10 beta software. The type 1 error rate and power are set at 5% and 80%, respectively.

Certainty for evidence

The certainty of the evidence will be assessed using GRADEpro, a tool developed by the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) Working Group [26]. The assessment criteria for certainty included, but are not limited to the initial study design, risk of bias, imprecision, indirectness, and inconsistency. Following the guidelines, the certainty of the evidence will be categorized as “High, Moderate, Low, or Very low” with the assistance of GRADEpro.

Patient and public involvement

No patient involved.

Ethics and dissemination

No ethical approval is required for this review. Our findings will be submitted to peer-reviewed journals.

Discussion

Although cycling exercise has demonstrated established benefits improving cardiopulmonary function and metabolic health in adult populations, its efficacy and safety profile in patients with CKD—particularly those undergoing dialysis—remain understudied and inconsistent. This knowledge gap is critically important given the unique physiological challenges in patients with CKD,

including accelerated muscle wasting, chronic inflammation, and fluctuating fluid balance, which may significantly influence exercise tolerance and therapeutic outcomes [20–24]. This meta-analysis will provide critical insights into the multidimensional effects of cycling interventions. Our findings will establish evidence-based thresholds for exercise intensity and duration tailored to CKD stages, addressing a long-standing clinical dilemma in balancing rehabilitation benefits with dialysis-related physiological constraints.

This review will adhere rigorously to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, with a pre-registered protocol (PROSPERO CRD420251048364) to mitigate selection bias. To enhance clinical applicability, we will incorporate a CKD-specific adaptation of the GRADE framework, emphasizing outcomes relevant to renal rehabilitation, such as changes in glomerular filtration rate (GFR), inflammatory biomarkers, and dialysis efficiency. Methodological robustness will be ensured through: (1) random-effects models accounting for heterogeneity in exercise protocols and CKD progression, (2) prespecified subgroup analyses stratified by dialysis status, baseline physical capacity, and comorbidity burden, and (3) sensitivity analyses confirming result stability after excluding studies with high attrition rates. These approaches will strengthen the validity of our conclusions across diverse clinical settings.

Clinically, our analysis will demonstrate that the therapeutic effects of cycling interventions in CKD management are closely tied to exercise intensity and clinical context. For hemodialysis patients, structured cycling sessions performed during dialysis treatment may be associated with measurable cardiovascular improvements and better fluid balance control, as evidenced by reduced interdialytic weight gain. However, excessively vigorous cycling regimens may be linked to elevated risks of procedure-related complications, including hypotensive episodes and exacerbated muscle fatigue. These findings contest the utility of generalized exercise guidelines and highlight the imperative for individualized protocols that account for patients' residual kidney function, hemodynamic profiles, and dialysis-specific physiological stressors.

Our review also has some limitations. This protocol acknowledges several potential limitations that may arise during the implementation of the meta-analysis. First, heterogeneity across eligible trials maybe expected to pose challenges, particularly regarding variations in cycling intervention modalities (e.g., in-dialysis versus home-based regimens), session durations, and outcome measurement tools. These discrepancies may introduce confounding effects that will

require careful statistical handling. Second, the scarcity of long-term follow-up data in existing literature may limit our ability to assess sustained therapeutic effects and delayed risks in advanced CKD populations. Third, methodological inconsistencies such as variable blinding practices and disproportionate representation across CKD stages (particularly underrepresentation of stage 4–5 non-dialysis patients) maybe predicted to affect result generalizability. Furthermore, potential publication bias will be systematically evaluated using funnel plots and Egger's regression. The absence of individual patient data (IPD) will be anticipated to restrict adjustments for critical covariates including nutritional status fluctuations and concomitant medication regimens.

To proactively address these limitations, the following methodological safeguards and research priorities will be implemented: Standardized data extraction protocols will be developed to harmonize cycling parameters (intensity, frequency, duration) with CKD progression stages and dialysis modalities. Core outcome measures will be predefined to include renal function trajectories, body composition metrics, and cardiovascular endpoints. Mechanistic subanalyses will be conducted where feasible, incorporating biomarker data (e.g., IL-6, TNF- α levels) to explore exercise-induced modulation of oxidative stress pathways. For studies reporting incomplete datasets, sensitivity analyses will be performed to evaluate the robustness of pooled effect estimates.

Abbreviations

RCTs	Randomized controlled trials
TSA	Trial sequential analysis
ROB 2	Revised Cochrane risk-of-bias tool
RevMan	Review Manager
RIS	Required information size
RR	Risk ratio
CI	Confidence interval
MD	Mean difference
PRISMA-P	Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocols
GRADE	Grading of Recommendations Assessment, Development, and Evaluation
CKD	Chronic kidney disease

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13643-025-02932-x>.

Additional file 1: PRISMA-P 2015 checklist.

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Not applicable.

Patient and public involvement

Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Provenance and peer review

Not commissioned; externally peer reviewed.

Authors' contributions

Chen Chen and Ji-ming Tao conceptualized and designed the study. Chen Chen drafted the manuscript. Chen Chen and Shao-hua Chen retrieved relevant databases. Chen Chen and Shao-hua Chen conducted literature screening and data extraction. Chen Chen and Shao-hua Chen assessed the risk of bias. Chen Chen and Yan Cao finished the table. Chen Chen, Shao-hua Chen, and Ji-ming Tao proofread the manuscript and improved the English writing. Chen Chen is responsible for the overall content as guarantor. All authors read and approved the manuscript.

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Data availability

All data and materials related to this systematic review and meta-analysis are available from the corresponding author upon reasonable request.

Declarations**Ethics approval and consent to participate**

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing financial interests or personal relationships that may have influenced the work reported in this study.

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